

Flashcard Learning System

Report Overview

Project Name	RecallIT
Document Type	User Interaction Test Suite
Scope	Authentication, Study Session Behavior, Progress
Total Cases	29 Test Cases
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Abstract

This document presents the RecallIT user interaction level test suite. The test suite validates major user-facing workflows, including authentication, study session behavior, card review rating, filter handling, daily progress tracking, and analytics visibility. Each test case includes ordered interaction steps and an expected result to support implementation verification and quality assurance.

Introduction

RecallIT is a flashcard-based learning system designed around authentication, deck-based study, review rating, spaced-repetition scheduling, and progress analytics. This test suite focuses on validating behavior from the user's perspective. The goal is to confirm that common workflows, invalid input states, edge cases, and progress-related UI updates behave consistently.

Test Suite Summary

Feature	Area Covered	Test Cases
1	Authentication and session behavior	8
2	Study filters, review rating, card flow, and empty-result behavior	12
3	Daily tracking, goals, heatmap, XP, streaks, forecast, and settings impact	9
Total	User interaction test coverage	29

Testing Approach

The test suite uses black-box, user-interaction-level validation. Each case is written as a sequence of actions followed by a clearly defined expected result. The tests are suitable for manual QA execution, UI acceptance testing, and conversion into automated end-to-end tests.

Feature 1: Authentication (Login Flow)

Scope: This feature contains 8 user-interaction test cases covering core work-flows, validation behavior, and user-facing results.

Test Case #1: Login with valid credentials

Test Steps

- a. Enter valid username/email
- b. Enter valid password
- c. Press login button

Expected Result: User logs in successfully and reaches the main study area.

Test Case #2: Login with invalid password

Test Steps

- a. Enter valid username/email
- b. Enter invalid password
- c. Press login button

Expected Result: Login is rejected and user sees an error message.

Test Case #3: Login with empty fields

Test Steps

- a. Keep username/email empty
- b. Keep password empty
- c. Press login button

Expected Result: Form validation blocks submission and prompts user to fill required fields.

Test Case #4: Register new user with valid data

Test Steps

- a. Switch to register mode
- b. Enter valid name, email, and password
- c. Press create account button

Expected Result: Account is created and user is logged into the main study area.

Test Case #5: Register with existing email**Test Steps**

- a. Switch to register mode
- b. Enter an email that already exists
- c. Press create account button

Expected Result: Registration is rejected with duplicate-account message.

Test Case #6: Login with invalid email format**Test Steps**

- a. Enter invalid email format
- b. Enter any password
- c. Press login button

Expected Result: Form validation prevents submission or server rejects request.

Test Case #7: Logout flow**Test Steps**

- a. Login successfully
- b. Press logout button
- c. Try opening protected study actions

Expected Result: Session ends and user is returned to auth screen.

Test Case #8: Session persistence after refresh**Test Steps**

- a. Login successfully
- b. Refresh browser page
- c. Wait for app initialization

Expected Result: User remains logged in and previous session is restored.

Feature 13: Study Session Behavior (Filters + Review Rating)

Scope: This feature contains 12 user-interaction test cases covering core workflows, validation behavior, and user-facing results.

Test Case #9: Start study with default behavior**Test Steps**

- a. Select a deck
- b. Press start study without applying extra filters
- c. Reveal answer and rate card

Expected Result: A due card is shown and rating action is accepted.

Test Case #10: Study with messy tags input**Test Steps**

- a. Enter tags with mixed case and duplicates (example: Cpp, cpp, Memory)
- b. Apply filters
- c. Start study

Expected Result: App still loads matching cards correctly (tags handled consistently).

Test Case #11: Rate card as Easy**Test Steps**

- a. Open a due card
- b. Reveal answer
- c. Press Easy

Expected Result: Review is saved, XP is awarded, and next-card/next-schedule behavior updates.

Test Case #12: Rate card as Good/Hard/Again**Test Steps**

- a. Open a due card
- b. Reveal answer
- c. Press one of: Good, Hard, or Again

Expected Result: Review is saved and progression reflects selected difficulty.

Test Case #13: Toggle answer visibility**Test Steps**

- a. Open a due card
- b. Click/tap to reveal answer
- c. Click/tap again to hide answer

Expected Result: Card flips between question and answer without losing card state.

Test Case #14: Scope filter set to Overdue

Test Steps

- a. Select due scope as **Overdue**
- b. Start study
- c. Observe returned cards

Expected Result: Returned cards are from overdue pool only.

Test Case #15: Scope filter set to Today

Test Steps

- a. Select due scope as **Today**
- b. Start study
- c. Observe returned cards

Expected Result: Returned cards are only those due within current day range.

Test Case #16: Weak-only filter enabled

Test Steps

- a. Enable **Weak only** checkbox
- b. Start study
- c. Review first few returned cards

Expected Result: Returned cards are from weak-card set and non-weak cards are excluded.

Test Case #17: Switch mode to Cram

Test Steps

- a. Set study mode to **Cram**
- b. Start study
- c. Continue multiple next cards

Expected Result: Queue includes urgent and near-term cards as per cram behavior.

Test Case #18: Switch mode to Exam

Test Steps

- a. Set study mode to **Exam**
- b. Start study
- c. Continue multiple next cards

Expected Result: Queue prioritizes unstable/risky cards per exam behavior.

Test Case #19: No cards match selected filters

Test Steps

- a. Apply very restrictive filters (tags + scope + weak-only)
- b. Start study
- c. Wait for response

Expected Result: User sees no-matching-cards state instead of app error.

Test Case #20: Add card then resume study

Test Steps

- a. Open add-card flow and create a valid card
- b. Return to study view
- c. Fetch next card

Expected Result: New card is available for study according to active filters.

Feature 29: Progress & Analytics (Daily Tracking + Heatmap)

Scope: This feature contains 9 user-interaction test cases covering core work-flows, validation behavior, and user-facing results.

Test Case #21: Daily review count updates

Test Steps

- a. Complete multiple reviews in one day
- b. Open metrics panel
- c. Check daily progress values

Expected Result: Reviewed count increases correctly for the current day.

Test Case #22: Goal progress visibility

Test Steps

- a. Set a daily goal
- b. Complete reviews toward that goal
- c. Refresh/open metrics

Expected Result: Remaining count and goal completion status update correctly.

Test Case #23: Heatmap range and day grouping

Test Steps

- a. Open analytics/heatmap view
- b. Inspect displayed date range
- c. Verify review activity appears on correct day cells

Expected Result: Heatmap spans the expected day window and groups activity by day consistently.

Test Case #24: Daily goal reached state

Test Steps

- a. Set a small daily goal (example: 2)
- b. Complete enough reviews to hit the goal
- c. Open progress section

Expected Result: Goal-met status becomes true and remaining count becomes zero.

Test Case #25: XP total increases after reviews

Test Steps

- a. Note current XP total
- b. Complete one or more rated reviews
- c. Recheck XP in UI/metrics

Expected Result: Total XP increases by expected reward amounts.

Test Case #26: Streak continuation on consecutive days

Test Steps

- a. Meet daily goal on day 1
- b. Meet daily goal again on day 2
- c. Open metrics/stats

Expected Result: Current streak increments by one and best streak updates

if needed.

Test Case #27: Forecast data visibility**Test Steps**

- a. Open metrics panel
- b. Locate due forecast section
- c. Inspect upcoming buckets

Expected Result: Forecast list is visible with non-negative due counts.

Test Case #28: Metrics reload behavior**Test Steps**

- a. Open metrics panel
- b. Trigger refresh action
- c. Observe panel state

Expected Result: Metrics reload successfully without losing current user context.

Test Case #29: Settings update affects progress display**Test Steps**

- a. Change daily goal in settings
- b. Save settings
- c. Re-open progress/metrics

Expected Result: Progress calculations reflect the updated daily goal.

Conclusion

The RecallIT test suite covers the system's primary user-facing workflows and validates both successful and failure-oriented scenarios. Authentication cases verify login, registration, validation, logout, and session restoration. Study-session cases validate filters, review ratings, answer visibility, and no-result handling. Progress and analytics cases verify metrics, goals, XP, streaks, forecasts, reload behavior, and settings-driven updates. Together, these cases provide a practical QA baseline for verifying RecallIT at the user interaction level.